Business Data Structures Analysis

Individual Report

Company – EDF Energy Ltd.

Submitted by:

Rashmi Satish Gajbhiye

ABSTRACT

Through text mining, EDF Energy reviews offer important understandings into customer emotions, service quality, and financial results. Visual tools like word clouds and associations showcase recurring themes, highlighting the value of helpfulness and service quality in achieving satisfaction. Positive and negative sentiment analysis points to areas for improvement, emphasizing friendliness and professionalism in service delivery. The examination of varying profit margins further accentuates the importance of strategic changes to effectively manage financial performance and foster enduring success.

Integrating a solid marketing strategy into the business blueprint is vital for elevating brand positioning, promoting engagement, and nurturing long-term success and loyalty. Capitalizing on text mining findings to refine financial outcomes, operational efficacy, and customer service will enhance EDF Energy's standing in the pursuit of enduring growth and sustained market competitiveness.

Contents

1. Introduction..... **Error! Bookmark not defined.**
2. Text Mining..... ...1
   2.1 Literature review on Text Mining 1
   2.2 Methodology...……..…….....…2
3. Text mining Analysis. 2
   3.1. Word Cloud.……....…..3
   3.2. Word Links ……...….4
   3.3. Common tokens and their frequencies……...5
   3.4. Word Trend……6
   3.5. Sentiment Analysis……6
   3.6. Thematic Analysis……..8
   3.7. Profit Margin……..9
4. Business Model ……9
5. Result and Discussion ……10
6. Recommendation……10
7. Conclusion……11
References 12
Appendices……...14

List of Figures:

Figure 1: Word Cloud using Voyant (Trustpilot, 2024)..... 3
Figure 2: Word Link using Voyant (Trustpilot, 2024)..... 4
Figure 3: Most Frequently Occurring Tokens Visualized using Excel (TrustPilot, 2024). 5
Figure 4: Word Trend formed using Voyant (TrustPilot, 2024)..... 6
Figure 5: Count of Sentiment formed on Excel (TrustPilot, 2024). 7
Figure 6: Sentiment Analysis from Excel (TrustPilot, 2024)…….....7
Figure 7: Themes Occurring in Comments Visualised on Excel (TrustPilot, 2024)……8
Figure 8: Profit Margin (Source: FAME; 2023)……...……9

1. Introduction:

EDF, the foremost carbon-free energy generator in the nation, is diligently fostering the growth of its indigenous nuclear, solar, and wind power assets. This strategic move intends to lessen the need for external energy supplies and shield against potential price escalations, ultimately bolstering energy autonomy and dependability (*About EDF*, 2024). EDF's operations in the UK encompass EDF Energy Holdings Limited alongside its subsidiary entities. Their core activities involve power generation from an array of sources such as nuclear, coal, gas, and renewables. Furthermore, these companies supply gas and electricity to a diverse customer base, comprising domestic, commercial, and industrial users (*Financial Information and Reports*, 2024).

EDF harnesses text mining, machine learning, and other data analytics approaches to bolster wind energy research. Text mining is applied to analyze wind turbine operation and maintenance reports, pinpointing failure-related terms and phrases to improve wind energy production and tackle operational hurdles (Ertek & Kailas, 2021). The central hypothesis posits that using automated text mining as a new, complementary technique for O&M data analysis can help predict potential failures across various types of machinery, enhancing operational reliability and efficiency (Blanco-M et al., 2019).

2. Text Mining:

2.1 Literature review on Text Mining:

Online platforms host diverse textual content spanning press releases, scientific publications, social media content, news articles, and customer feedback. Management researchers exploit text mining capabilities to build theories across multiple domains, capitalizing on the expansive, high-quality unstructured data arising from widespread digital transformation. This wealth of unstructured data enables researchers to employ big data analytics tools like text mining for analysis and valuable insight extraction. Thus, theory-building methodologies integrating text analytics on substantial unstructured data sets are steadily gaining prominence (Kar & Dwivedi, 2020).

Text mining's significance continues to grow within the context of service management as an increasing number of individuals tap into extensive data resources hosted on digital platforms that facilitate these services, emphasizing its relevance and practicality in this field (Kumar et al., 2021). Utilizing text mining allows for the extraction of valuable information from various input documents, efficiently pinpointing key findings within substantial quantities of pre-processed textual data.

With its outstanding capacity to scrutinize unstructured data, text mining is widely employed in diverse fields, including political document assessment, IT, energy and service sectors, as well as healthcare services (Chu et al., 2020).

2.2 Methodology:

- **Data Collection:** Octoparse, a web scraping tool, was employed to gather EDF Energy customer feedback from Trustpilot. The extracted reviews were stored in an Excel spreadsheet. This data collection step was crucial in preparing for sentiment analysis to assess customer opinions about EDF Energy based on the collected feedback.
- **Text Pre-processing:** Data cleaning and formatting were accomplished through text pre-processing, which involved transforming text into lowercase, excluding stop words and special characters, and removing blank rows and columns. These procedures ensured consistent data formatting while eradicating unneeded or distracting elements from the dataset. Text pre-processing methods impact the performance of machine learning techniques in automatically classifying work orders and prioritizing maintenance requests (Shamshiri et al., 2024).

- **Application of text mining tools:** NVivo for Thematic Analysis was used to import collected reviews, coding them by themes, dates, and reasons, and generating graphical representations. Voyant was employed to create a word cloud, word frequency associations, and relative frequency trends from pre-processed reviews, considering stop word removal. Lastly, Excel's Azure Machine Learning facilitated sentiment analysis.

3. Text mining Analysis:

A dataset comprising 680 Trustpilot reviews of EDF Energy were gathered to conduct a text-mining analysis for the EDF Energy (Trustpilot, 2024). The findings, obtained through thorough analysis, provide valuable insights into customer feedback and experiences with the company.

3.1. Word Cloud:

Figure 1: Word Cloud using Voyant (Trustpilot, 2024)

In Figure 1, the word cloud visually vividly displays the diverse sentiments expressed in the comments, aligning with the sentiment analysis outcomes. The customer service word cloud graphically depicts key components and experiences in the industry. Larger words like "helpful," "service," "customer," and "thank" reflect their significance and prevalence, stressing the value of support, commitment, and appreciation in establishing strong customer connections. Smaller terms such as "problem," "account," "issue," "team," and "polite" offer additional context, indicating challenges, the need for teamwork, and the significance of maintaining respectful interactions. Together, these elements illustrate the complexity of customer service and emphasize the importance of providing exceptional customer experiences.

The word cloud underscores positivity and outstanding customer service as key themes. Words like "thank you," "helpful," and "excellent" show the importance of gratitude and delivering superior service. This aligns with customer service's primary aim of fostering satisfaction and strong bonds through proactive help and positive interactions. By visually representing core concepts and goals, the word cloud emphasizes the significance of nurturing pleasant experiences and customer relationships.

3.2. Word Links:

Figure 2: Word Link using Voyant (TrustPilot, 2024)

In Figure 2, the word link visualization connects various customer service elements based on feedback. The core theme revolves around three essential elements: "Helpful," "Friendly," and "Customer." "Helpful" represents a supportive and collaborative environment where individuals work together effectively, providing timely support and appreciating each other's efforts. "Friendly" brings together professionalism and warmth, encouraging long-term friendly and approachable interactions. Together, these nodes contribute to an inviting and collaborative setting.

The "Customer" node accentuates key factors for customer satisfaction, such as good manners, top-notch service, and positive experiences. It emphasizes a customer-centric mind-set and the importance of fulfilling or exceeding customer expectations. While "Helpful," "Friendly," and "Customer" form the main theme, other concepts like "energy," "excellent," "issue," "skills," and "sort" might add more depth, despite their unclear relationships. Together, these elements create a complex theme network that embodies a welcoming, cooperative, and customer-focused atmosphere.

3.3. Common tokens and their frequencies:

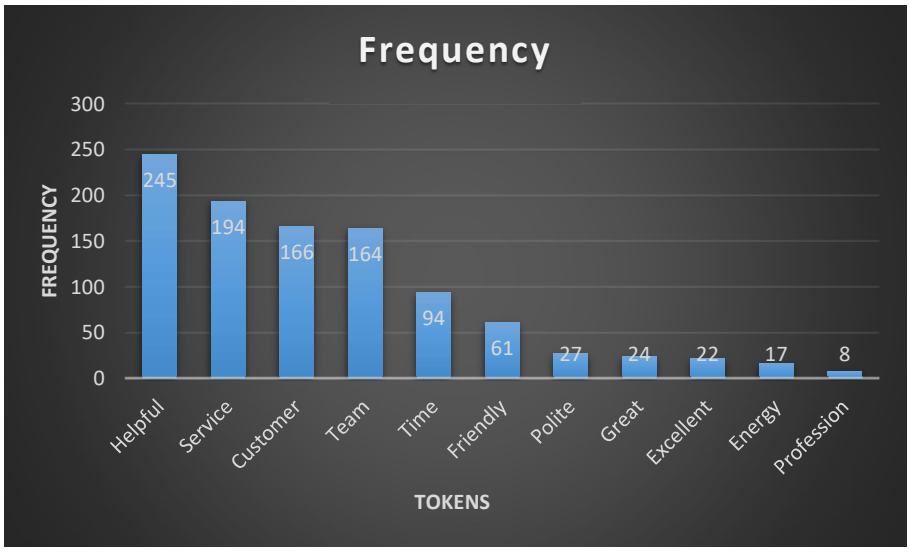

Figure 3: Most Frequently Occurring Tokens Visualized using Excel (TrustPilot, 2024)

In Figure 3, the graph presents a clear visual illustration of token instances within customer support. Key customer service aspects like "Helpful," "Service," and "Customer" are positioned on the x-axis marked "TOKENS." The y-axis shows "FREQUENCY," with numbers from 0 to 300 indicating token frequency counts. Blue bars represent token frequencies, e.g., "helpful" appears 245 times, "service" 194 times, and "customer" 166 times. Tokens such as "Team," "Time," "Friendly," "Polite," and "Great" impact the varying frequencies in the graph.

The graph's main theme likely focuses on the significance of positive qualities in customer service. Implied terms such as "great," "helpful," and "excellent" highlight the importance of attributes associated with outstanding customer experiences. This theme aligns with the depicted frequency counts, stressing the value of excellence, support, and appreciation in building solid customer connections. The graph visually represents key tokens' frequencies, underlining the vital role of positive traits in enhancing customer satisfaction and loyalty within the realm of customer service.

3.4. Word Trend:

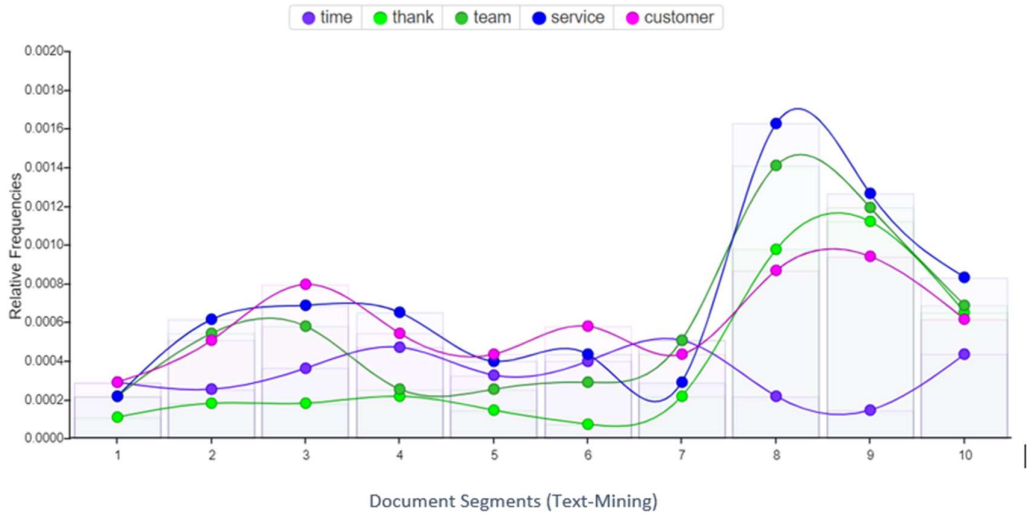

Figure 4: Word Trend formed using Voyant (TrustPilot, 2024)

The line plot showcases relative frequencies of five phrases from text-mining analysis across ten segments. Each phrase's fluctuating frequency indicates its relevance within the text data. "Customer" surges in segment 9, highlighting significant focus in that part. "Service" stays steady, suggesting consistent importance. "Thank" has moderate frequency, indicating presence but less emphasis than "Customer" or "Service." The lower frequencies and instances of "Time" and "Team" imply reduced significance compared to other phrases.

The graph visualizes term frequencies, signifying their relative significance in the text under analysis. "Customer" peaking in segment 9 implies a crucial point or discussion on customer-centric matters. Alongside consistent "Service" frequency and recurring "Thank" instances, the text's core theme revolves around successful customer interactions and teamwork. The graph identifies both frequency patterns and primary themes, shedding light on significant elements and focal points within the text.

3.5. Sentiment Analysis:

The objective of sentiment analysis is to extract and analyze the emotions present in textual data. This technique is employed in various contexts, such as surveys, product reviews, and

customer feedback, to gauge public opinion. Sentiment analysis also plays an essential role in crucial tasks like managing public perception, social media surveillance, and improving customer satisfaction.

Row Labels	Count of Sentiment
negative	288
neutral	92
positive	300
Grand Total	**680**

Figure 5: Count of Sentiment formed on Excel (TrustPilot, 2024)

The table classifies sentiments into three groups: Positive, Neutral, and Negative. Negative sentiments, numbering 288 instances, express dissatisfaction or criticism. Neutral sentiments, with 92 instances, signify a lack of strong emotion or ambiguity. Positive sentiments are most common, with 300 instances indicating satisfaction and approval. In total, 680 sentiments across categories reflect varied perspectives and emotions within the data set.

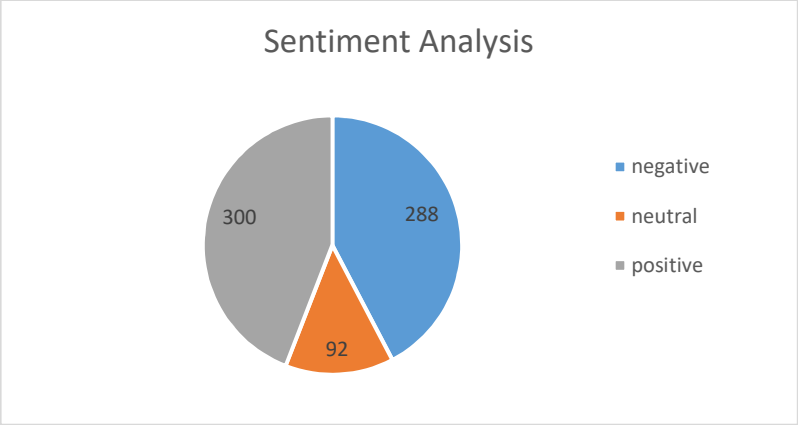

Figure 6: Sentiment Analysis from Excel (TrustPilot, 2024)

The pie chart displays sentiment ratios across Negative, Neutral, and Positive categories. Positive sentiments represent the smallest share, while Negative sentiments dominate, closely followed by Neutral. This highlights the importance of understanding and addressing negative

sentiments to achieve better sentiment balance. Examining sentiment ratio patterns over time or within different datasets provides businesses with insights into shifting customer attitudes and market dynamics, enabling data-driven decisions and targeted improvements in customer engagement strategies, ultimately boosting customer satisfaction.

3.6. Thematic Analysis:

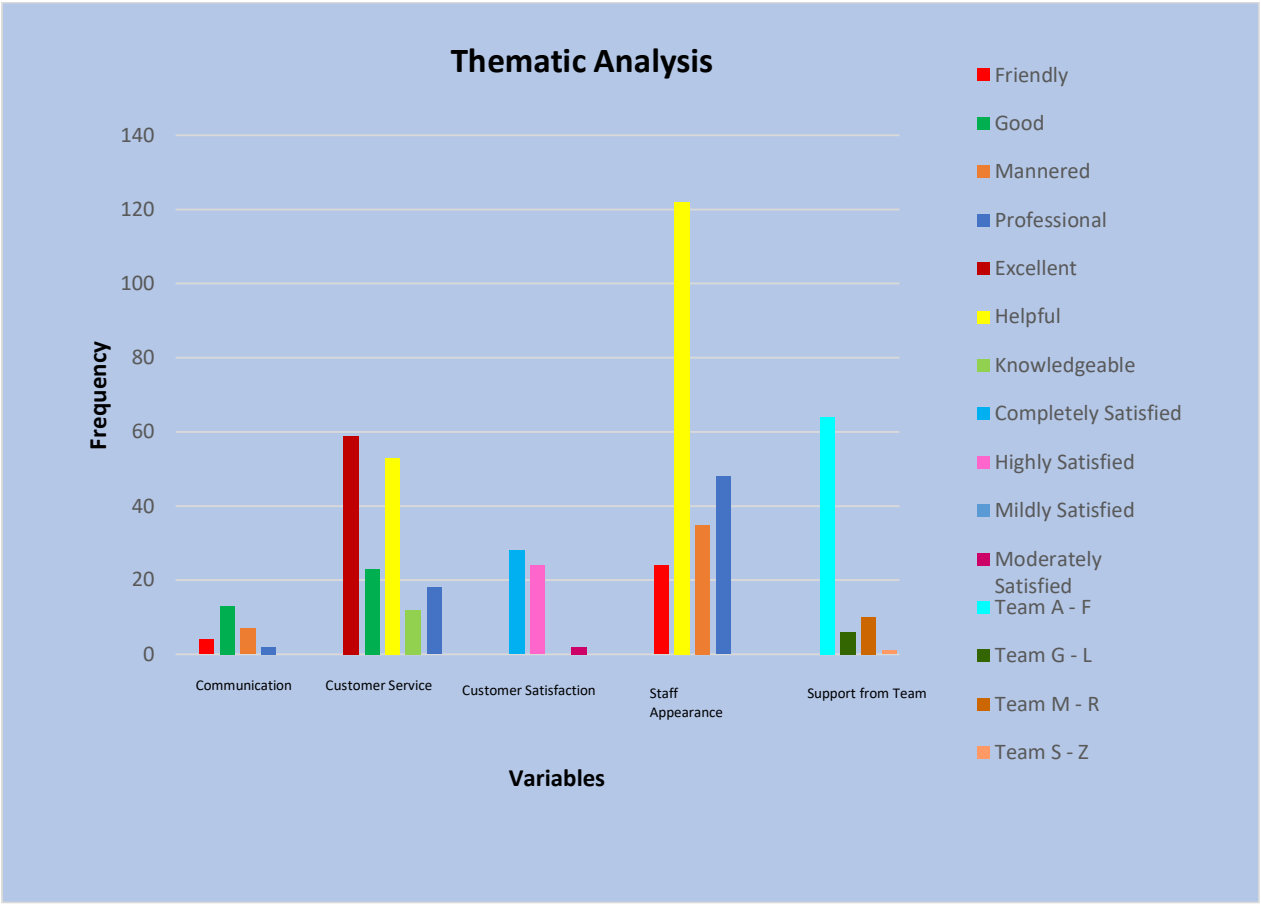

Figure 7: Themes Occurring in Comments Visualised on Excel (TrustPilot, 2024)

The thematic analysis chart examines positive customer service using variables and related themes. The chart displays variables like Communication, Customer, Staff Appearance, and Support with themes including Excellent, Professional, and Friendly. Theme frequencies within variables offer insights into sentiment distribution. Significantly, the 'Excellent' theme prevails in the 'Support from Team A - F' variable, suggesting strong positive sentiment. The

chart provides an extensive overview of sentiment across service dimensions, enabling better understanding and improving customer satisfaction and service quality.

3.7. Profit Margin:

Figure 8: Profit Margin (Source: FAME; 2023)

The "Profit Margin" chart illustrates profit margin fluctuations between 2018 and 2022. Years are plotted on the x-axis, while the y-axis ranges from 0 to -10. A blue line graph traces the evolving profit margin, beginning near 0, declining sharply to nearly -9 in 2019, gradually recovering to -3 in 2020, then dipping to -5 in 2021, and finally bouncing back to about -3 in 2022. These shifts highlight significant financial performance volatility, suggesting a possible rebound in 2020 followed by additional hardships. Monitoring profit margins closely and implementing strategic changes will be crucial for EDF Energy Ltd. to handle these fluctuations successfully.

4. Business Model:

EDF Energy ranks among reliable UK energy providers. Despite appearing ideal for small business, leveraging Bionic's comparison services can help ensure that customer access the most competitive offers in the market (EDF Energy, Compare Business Energy Suppliers,

Bionic, 2023). To achieve maximum gross margin and security, EDF meticulously optimizes its asset portfolio at each stage of power generation and sales (*Optimisation and Trading*, 2022b).

A market strategy is very important to add in business model. An exhaustive marketing plan generally includes vital elements such as brand objectives, target audience characteristics, marketing channels, key performance indicators, and other pertinent aspects to facilitate successful implementation and evaluation of marketing campaigns (Lauron, 2024). Developing a concise plan is crucial to create a governance structure that supports strategic goals and delineates roles and responsibilities, ensuring smooth and efficient implementation (*Optimizing Your Organization's In-house Legal Operating Model*, 2023).

5. Result and Discussion:

Comprehensive text mining of 680 Trustpilot reviews for EDF Energy provides valuable understandings about customer feelings, service experiences, and financial results. The word cloud visualization emphasizes positive engagement themes, like service quality and customer gratitude, which are vital for satisfaction. Word associations further highlight the significance of cooperative, customer-oriented approaches in delivering services.

The sentiment analysis, however, reveals prevalent negative emotions, signalling EDF Energy must address dissatisfaction causes to enhance sentiment balance and improve satisfaction. Thematic analysis digs into essential service dimensions, focusing on quality, professionalism, and friendliness. A fluctuating profit margin chart indicates unstable financial performance, stressing the importance of sentiment monitoring, service quality enhancement, and strategic adaptation to financial changes while preserving customer satisfaction and loyalty.

6. Recommendation:

1. Predictive analytics: Utilizing customer data for trend prediction and service customization fosters increased customer satisfaction and loyalty by offering personalized experiences and showing recognition of individual customer requirements.

2. Feedback Mechanism: Implement a streamlined system for capturing and evaluating customer feedback, allowing real-time modifications to products or services to address issues promptly and continuously enhance service quality.

3. Customer Care Initiative: To tackle the uptick in negative feedback, create a customer care program focused on addressing primary concerns swiftly and effectively, potentially by assembling a specialized customer resolution team to improve satisfaction by promptly addressing and resolving customer problems.

7. Conclusion:

Text mining of EDF Energy reviews provides key insights into customer emotions, service quality, and financial results. Visuals like word clouds and associations accentuate the value of helpfulness and service quality for customer satisfaction by showcasing recurring themes. Negative sentiments, however, indicate areas for improvement to boost overall sentiment. Thematic analysis highlights professionalism and friendliness as crucial service aspects, while fluctuating profit margins underscore the need for strategic changes to adeptly manage financial performance. Integrating a solid marketing strategy into the business plan is vital for driving engagement, strengthening brand positioning, and fostering enduring success and loyalty.

References:

1. Kumar, S., Kar, A. K., & Ilavarasan, P. V. (2021). Applications of text mining in services management: A systematic literature review. *International Journal of Information Management Data Insights*, *1*(1), 100008. https://doi.org/10.1016/j.jjimei.2021.100008
2. Kar, A. K., & Dwivedi, Y. K. (2020). Theory building with big data-driven research – Moving away from the "What" towards the "Why." *International Journal of Information Management*, *54*, 102205. https://doi.org/10.1016/j.ijinfomgt.2020.102205
3. Chu, C., Park, K., & Kremer, G. E. (2020). A global supply chain risk management framework: An application of text-mining to identify region-specific supply chain risks. *Advanced Engineering Informatics*, *45*, 101053. https://doi.org/10.1016/j.aei.2020.101053
4. *An Interview with EDF Group: How EDF is using its raison d'être to transition to a new energy model*. (2024). ERM. https://www.sustainability.com/thinking/an-interview-with-edf-group/
5. *EDF Energy| Compare Business energy Suppliers | Bionic*. (2023). https://bionic.co.uk/business-energy/suppliers/edf-energy/
6. Lauron, S. (2024, March 29). How to Create a Complete Marketing Strategy in 2024 [Data + Expert Tips]. *https://blog.hubspot.com/marketing/marketing-strategy*. https://blog.hubspot.com/marketing/marketing-strategy
7. *Optimizing your organization's in-house legal operating model*. (2023, February 8). Deloitte. https://www.deloitte.com/global/en/services/legal/perspectives/in-house-legal-operating-model.html
8. *Financial information and reports*. (2024). EDF. https://www.edfenergy.com/about/financial-information
9. *Fame*. (2023). https://fame-r1.bvdinfo.com/version-20240321-1-1/fame/1/Companies/report/Index?backLabel=Back%20to%20Profit%20%26%20loss%20account&format=_standard&BookSection=PROFILE&seq=0&sl=1712157640993

10. *Optimisation and trading*. (2022b, July 21). EDF FR. https://www.edf.fr/en/the-edf-group/edf-at-a-glance/optimisation-and-trading

11. Ertek, G., & Kailas, L. (2021). Analyzing a Decade of Wind Turbine Accident News with Topic Modeling. *Sustainability (Basel)*, *13*(22), 12757. https://doi.org/10.3390/su132212757

12. Blanco-M, A., Martí-Puig, P., Gibert, K., Cusidó, J., & Solé-Casals, J. (2019). A Text-Mining approach to assess the failure condition of wind turbines using maintenance service history. *Energies*, *12*(10), 1982. https://doi.org/10.3390/en12101982

Appendices

1. https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7B9DE62288-D0DE-4802-9A67-6A46E834E42C%7D&file=Comments.xlsx&action=default&mobileredirect=true

2. https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7B202BB7CA-85D6-45FD-AE51-F449D623A2B1%7D&file=Word%20Frequency.xlsx&action=default&mobileredirect=true

3. https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7B2FEC259A-3CC8-4805-8F70-DC23D2DEE070%7D&file=Sentiment%20Analysis%201.xlsx&action=default&mobileredirect=true

4. https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7B6C0E11A6-2B19-4B95-AAEF-F155210B9A6A%7D&file=Profit%20Margin.xlsx&action=default&mobileredirect=true

5. https://livecoventryac-my.sharepoint.com/:x:/r/personal/gajbhiyer_uni_coventry_ac_uk/_layouts/15/Doc.aspx?sourcedoc=%7BBD6E697D-DCEA-415A-B14C-534306112493%7D&file=Thematic%20Analysis.xlsx&action=default&mobileredirect=true